Early and Late Fusion of Frozen Foundation Models for IMU-Based Human Activity Recognition

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Abstract

This work presents the UT-IR team's submission to the Sussex–Huawei Locomotion (SHL) Challenge 2026, conducted under a strict evaluation framework that requires all foundation models to remain frozen. To achieve robust transportation mode recognition across heterogeneous user data, we systematically investigate and integrate five diverse time-series foundation models spanning both classification- and forecasting-oriented pretraining paradigms: Mantis, TSPulse, UniShape, NuTime, and TiRex. We design lightweight, task-specific classification heads—including a custom multi-branch gating network that arbitrates between deep temporal embeddings, spectral features, and statistical heuristics—to adapt these frozen representations. Furthermore, we extensively evaluate both early (feature-level cross-attention) and late (decision-level) fusion paradigms to synergize the complementary strengths of the constituent models. Our experimental results demonstrate that while high-capacity non-linear meta-learners achieve near-perfect validation accuracy, a mathematically bounded, combinatorially optimized global linear fusion strategy provides superior robustness against domain shifts, yielding a highly stable 5-model consensus architecture for final deployment.

CCS Concepts

• **Computing methodologies** → **Neural networks**; **Ensemble methods**.

Keywords

Human Activity Recognition, Transportation Mode Recognition, Time-Series Foundation Models, Fusion Strategies, Time-Series Representation Learning, Ensemble Learning

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1 Introduction

Human locomotion activity recognition is a fundamental task in ubiquitous computing, enabling a wide range of applications, including personalized navigation, context-aware mobile services (e.g., automatic transportation mode switching), and long-term health monitoring for chronic disease management and patient follow-up. Although the widespread adoption of wearable devices and smartphones has facilitated continuous motion sensing, acquiring large-scale datasets that capture natural human behavior in unconstrained, real-world environments remains a significant challenge. To address this challenge, the Sussex-Huawei Locomotion (SHL) dataset [5, 10] was introduced as a large-scale benchmark, comprising seven months of continuous smartphone sensor recordings collected from 3 users in everyday environments. Since its release,

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the annual SHL recognition challenge has provided a standardized benchmark for advancing transportation mode recognition using this dataset.

The SHL recognition challenge 2026 introduced a new evaluation protocol centered on frozen foundation models (FMs), requiring all pretrained model parameters to remain fixed throughout the entire recognition pipeline. Motivated by this setting, we investigate the effectiveness of leveraging multiple complementary foundation models (FMs) for transportation mode recognition. Specifically, we evaluate five diverse time-series foundation models spanning both classification- and forecasting-oriented pretraining paradigms: Mantis, TSPulse, UniShape, NuTime, and TiRex, and compare early- and late-fusion strategies for integrating their representations.

The remainder of this paper is organized as follows. Section 2 describes the proposed framework, Section 3 presents the experimental results and discussion, and Section 4 concludes the paper.

2 The Proposed Pipeline

An overview of the proposed inertial measurement unit (IMU)-based human activity recognition (HAR) pipeline, built upon frozen FMs, is illustrated in Figure 1. This section begins with a concise overview of the five FMs employed in the framework, followed by a detailed description of the proposed early and late fusion strategies used to combine their learned representations for robust activity recognition.

2.1 Preprocessing

The raw input data consists of 9-channel IMU signals, comprising tri-axial accelerometer, gyroscope, and magnetometer readings. As an initial step, the continuous data stream is segmented into 5-second non-overlapping windows to provide discrete, fixed-length inputs for the FMs. Each window is then band-pass filtered (0.3–20 Hz) to remove gravitational drift and high-frequency sensor noise, followed by instance normalization to achieve zero mean and unit variance. This normalization reduces subject- and sensor-specific variability while aligning the input distribution with the pre-training data of the frozen FMs, enabling more effective feature extraction.

2.2 Foundation Model Backbones

To construct a robust and generalized feature extraction pipeline for HAR, we employ five time-series FMs spanning different pre-training objectives and architectural families. Together, these models produce complementary embeddings that differ in tokenization strategy, representation granularity, and inductive bias.

Classification-Native Transformers. This group comprises models explicitly pre-trained for time-series classification using self-supervised or weakly supervised objectives:

- **MantisV2** [4] combines raw signals, first-order differences, and local statistical descriptors during tokenization, producing embeddings that capture both short-term motion dynamics and longer temporal trends.
- **NuTime** [6] employs Numerically Multi-scaled Embeddings that preserve signal amplitudes across multiple numerical scales, facilitating discrimination between activities with different motion intensities.
- **UniShape** [8] introduces a Shape-Aware Adapter that models discriminative local subsequences (shapelets), yielding interpretable representations well suited to fine-grained activity recognition.

Cross-Task Heterogeneous Models. To further diversify the representations, we additionally include two FMs originally developed for forecasting and diagnostics:

- **TiRex** [2] is a zero-shot forecasting model based on the xLSTM architecture, whose explicit state tracking captures long range temporal dependencies that may be less effectively represented by patch-based attention mechanisms.
- **TSPulse** [3] is an ultra-lightweight TSMixer-based model that disentangles temporal, spectral, and semantic representations, providing a complementary multi-view embedding space.

Owing to these heterogeneous embedding spaces, a single downstream classifier is unlikely to be optimal across all backbones. We therefore design lightweight backbone-specific classification heads while keeping all FM parameters frozen.

2.3 FM-Specific Head Design

For each frozen backbone, we define a structured design space for the late-fusion classification head. The design choices are organized into five orthogonal axes.

- **Cross-Channel Interaction.** For backbones producing channel-wise embeddings, inter-channel dependencies are modeled using one of three architectures: (i) *CLS-Token Transformers*, which prepend a learnable global token to the channel sequence and extract interactions via self-attention; (ii) *Seq-to-Seq Encoders*, which treat the flattened feature vector as a sequence of fixed-length tokens; and (iii) *Global Vector Processing*, which bypasses sequential modeling for dense projections.
- **Complementary Features.** Backbone embeddings may be augmented with frequency-domain and statistical descriptors to recover complementary information that may not be preserved during FM pre-training.
- **Readout.** The processed tokens are aggregated using either *Pooling*, *Concatenation*, or *Dynamic Gating*.
- **Regularization.** *Mixup* augmentation [12], *Label Smoothing* (LS) [9], and *Token Dropout* at both token and block levels are applied to reduce overfitting.
- **Objective.** Training is performed using either standard Cross-Entropy (CE), Focal Loss [7], or class-weighted objectives. Focal Loss is considered to mitigate class imbalance by down-weighting easy examples, while static class weights increase the contribution of underrepresented activities.

Selection Protocol. Candidate head configurations were evaluated on the validation split. Model selection was based on validation F1 score and accuracy. The final backbone-specific head configurations are summarized in Table 1. These backbone-specific heads provide adaptation layers that accommodate the heterogeneous embedding spaces produced by different FMs while preserving the frozen-backbone training paradigm.

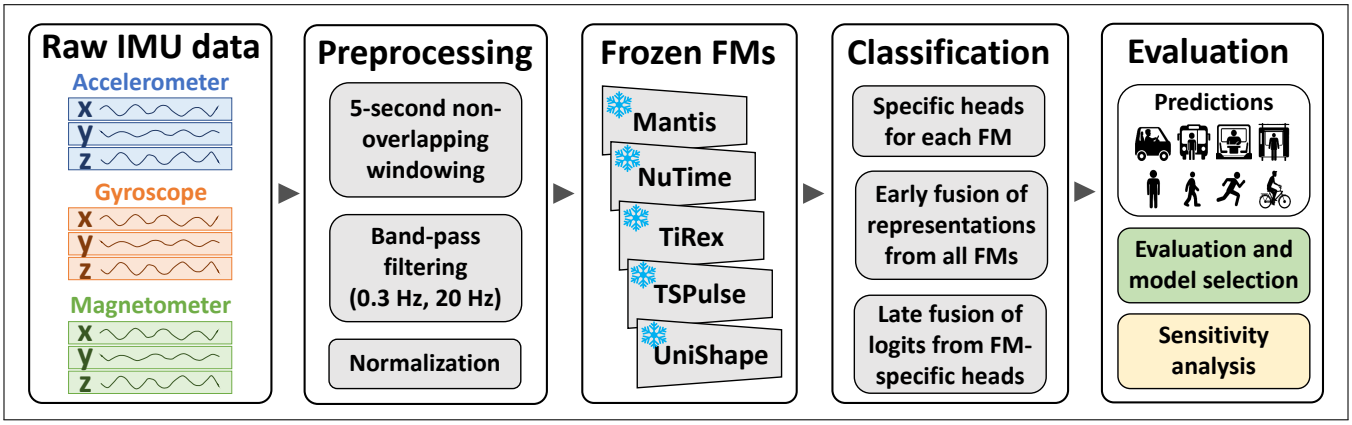


Figure 1: Overview of the proposed IMU-based HAR pipeline utilizing frozen foundation models (FMs).

Table 1: Final FM-specific head configurations.

Foundation Model	Cross-Channel Interaction	Complementary features	Readout	Regularization	Objective
MantisV2	CLS-Token Transformer	✗	Pooling	Mixup	CE
TSPulse	Seq-to-Seq	✗	Concatenation	Mixup, LS, Dropout	CE
UniShape	CLS-Token Transformer	✗	Pooling	Mixup	CE
NuTime	Seq-to-Seq	✓	Concatenation	Mixup, LS, Dropout	CE
TiRex	Global Vector Processing	✓	Dynamic Gating	Mixup, LS, Dropout	Focal + CW

Abbreviations: LS (Label Smoothing), CE (Cross-Entropy), CW (Class Weights).

2.4 Fusion Strategies

Motivated by the increasing interest in exploiting the complementary capabilities of diverse FMs, and encouraged by the recent study [1] demonstrating that ensembles of FMs enhance robustness to data variability, the proposed framework combines five distinct frozen FMs through two fusion paradigms: early fusion and late fusion. An overview of these fusion strategies is presented in Figure 2, while their architectures and implementation details are described in the following two subsections.

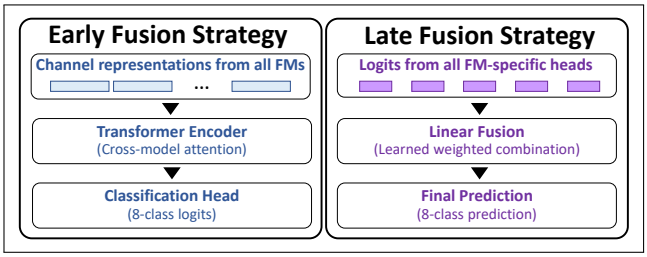


Figure 2: Early and late fusion strategies.

2.4.1 Early Fusion Ensemble Strategy. In the early fusion framework, the feature representations produced by the five FMs differ in both dimensionality and structural format. To enable their integration, the outputs of all models are first projected into a shared latent space through model-specific learnable linear projection layers. For the four sequence-based models (UniShape, Mantis, NuTime, and TiRex), learnable channel embeddings are subsequently incorporated into the projected representations to preserve the spatial relationships among sensor channels. In contrast, the representations generated by TSPulse, which lack a temporal sequence dimension and instead provide holistic feature embeddings, are projected directly into the shared latent space without the addition of channel embeddings.

To capture the complex interdependencies among the FMs, a specialized cross-attention mechanism is introduced for the sequence-based representations. Within this module, the output sequence of each of the four sequence-based FMs independently serves as the source for query generation, while the output sequences of the remaining three models are used to compute the corresponding keys and values. This design enables each model to selectively attend to complementary information learned by the other FMs, thereby promoting richer cross-model feature interactions. To improve robustness and reduce over-reliance on specific features during training, a stochastic token masking strategy is applied to the key representations. Following the cross-attention stage, a dynamic channel-attention pooling module computes adaptive importance weights for each channel, aggregating the sequence-level features into a compact, model-specific representation vector.

In the final stage, the attention-pooled vectors from the sequence-based FMs are concatenated with the projected TSPulse representation and fed into an MLP classification head. We explored alternative fusion strategies, such as prepending a global classification (CLS) token to the concatenated sequence of all model tokens. However, this approach rapidly overfitted the training data, required heavy

regularization, and failed to effectively leverage representations across all FMs. In contrast, our proposed lightweight ensemble explicitly guides cross-model interactions, effectively exploiting the heterogeneous FMs to yield a robust, unified representation for HAR.

2.4.2 Late Fusion Ensemble Strategy. Two late fusion strategies were investigated in this study. In both approaches, the raw output logits produced by each FM are first transformed into class probability distributions using the Softmax function. Let $P_m \in \mathbb{R}^{N\times C}$ denote the probability matrix for model m, where N is the number of samples and $C = 8$ is the number of transportation mode classes.

Class-Weighted Non-Linear Fusion (XGBoost): We first employed XGBoost as a non-linear meta-learner to capture complex, conditional dependencies between the FMs. To combat the severe class imbalance inherent in the SHL dataset, we applied a balanced sample weighting scheme during the gradient boosting phase. The weight for each sample belonging to class c was dynamically assigned as:

$$W_c = \frac{N_{total}}{C \times N_c} \tag{1}$$

where N_{total} is the total number of validation samples and N_c is the frequency of class c. This forced the meta-learner to heavily penalize misclassifications in minority classes rather than over-optimizing for majority classes.

Global Linear Fusion via Exhaustive Simplex Search: To provide a robust, variance-bounding alternative to tree-based meta-learning, we implemented a scalar-weighted linear ensemble. To guarantee convergence to the global empirical optimum and circumvent the local minima associated with gradient-based weighting, we conducted an exhaustive grid search over the standard $(M-1)$-simplex. Utilizing a discretization step size of 0.01 for $M = 5$ models, we generated the complete set of 4,598,126 valid weight permutations satisfying $\sum_{m=1}^{M} w_m = 1$. This combinatorial search space was evaluated through memory-efficient, highly parallelized batched tensor operations via Einstein summation on an NVIDIA GeForce RTX 3090 GPU:

$$\mathcal{E}_{p,n,c} = \sum_{m=1}^{M} \mathcal{W}_{p,m}\mathcal{P}_{n,m,c} \tag{2}$$

where $\mathcal{W} \in \mathbb{R}^{P\times M}$ is the matrix of all P generated weight permutations, $\mathcal{P} \in \mathbb{R}^{N\times M\times C}$ is the stacked validation probability tensor for N samples across M models and C transportation classes, and $\mathcal{E}$ is the resulting batched ensemble probability tensor. The optimal weight configuration was subsequently extracted by maximizing the macro F_1-score over the final predictions $\hat{Y}_{p,n} = \arg\max_c \mathcal{E}_{p,n,c}$.

3 Experimental Results and Discussion

All experiments were conducted on the SHL dataset [5, 10]. This dataset was collected from three volunteers carrying four smartphones positioned at the hand, hip, torso, and bag. Following the official SHL challenge protocol, the sensor data are segmented into non-overlapping 5-second windows, each consisting of nine sensor channels. The official user-disjoint partition includes 196,072 training windows from User 1, 28,789 validation windows from Users 2 and 3, and 92,726 temporally shuffled test windows from Users 2 and 3. Although the official protocol ensures user-disjoint evaluation, the training set contains data from only a single subject, which may lead to overfitting to user-specific motion characteristics and reduce generalization to unseen individuals. To mitigate this issue, we reorganized the dataset at the representation level by programmatically sampling a fixed 25% of the validation embeddings (28,789 samples) to serve as our new validation holdout, while merging the remaining 75% (86,367 samples) with the original training embeddings to form a robust 870,655-sample training corpus.

Table 2 presents a comparative evaluation of the five individual FMs and their ensembles, obtained using the proposed early and late fusion strategies, on the enriched training set and the held-out validation set.

Table 2: Performance comparison of various FMs and fusion approaches. Metrics reported include micro Accuracy (Acc) and macro F_1-Score (F_1). EF and LF denote Early Fusion and Late Fusion, respectively. All results are reported as percentages on the enriched training set and the held-out validation set.

Model	Training Set		Validation Set	
	Acc	F_1	Acc	F_1
MantisV2 (M)	99.68	99.71	91.44	91.89
NuTime (N)	98.52	98.62	**93.87**	**94.32**
TiRex (Ti)	92.17	92.79	91.26	91.87
TSPulse (Ts)	88.99	90.03	82.99	83.66
UniShape (U)	87.12	88.32	86.23	86.73
EF (M+N+Ti)	94.11	94.60	92.62	93.09
EF (U+M+N)	91.21	92.02	89.77	90.37
EF (U+M+N+Ti)	94.42	94.89	**93.08**	**93.55**
EF (U+M+N+Ti+Ts)	96.82	97.08	92.63	93.17
LF (Linear)[a]	–	–	94.66	95.18
LF (XGBoost)	–	–	**96.32**	**96.87**

[a]This model was selected for generating the final predictions on the unlabeled official test set of the SHL Recognition Challenge 2026.

3.1 Early Fusion Evaluation

As shown in Table 2, integrating all FMs (EF U+M+N+Ti+Ts) did not yield the highest validation performance. This outcome is likely due to the structural differences of the TSPulse tokens, which lack spatial channel dimensions, making alignment with sequence-based models difficult during early feature blending. The four-model configuration comprising UniShape, Mantis, NuTime, and TiRex (EF U+M+N+Ti) achieved the highest early fusion performance with 93.08% accuracy and a 93.55% macro F_1 score. A comparison of the triplet configurations, specifically EF (U+M+N) and EF (M+N+Ti), indicates that each base model provides non-redundant representations. The ensemble cannot be further ablated without performance loss, although TiRex contributes more to the overall predictive accuracy.

The cross-attention mechanism serves as an effective routing tool to exploit the complementary characteristics of the heterogeneous FMs. Figure 3 illustrates the four attention heads within the

single cross-attention layer of the EF (U+M+N+Ti) ensemble, where the tokens of each querying FM (y-axis) attend to the tokens of the other models (x-axis). As depicted, the attention weights exhibit distinct patterns that vary significantly not only between models but also across the four independent heads. This head-specific diversity indicates that each querying model learns to project the features of its peers into multiple distinct representational subspaces, allowing it to simultaneously extract different types of contextual information. By capturing these interdependencies, this lightweight cross-model interaction generates robust features for classification. Consequently, the ensemble avoids the severe training set overfitting observed in the standalone Mantis and NuTime models, which rely on relatively heavy self-attention mechanisms.

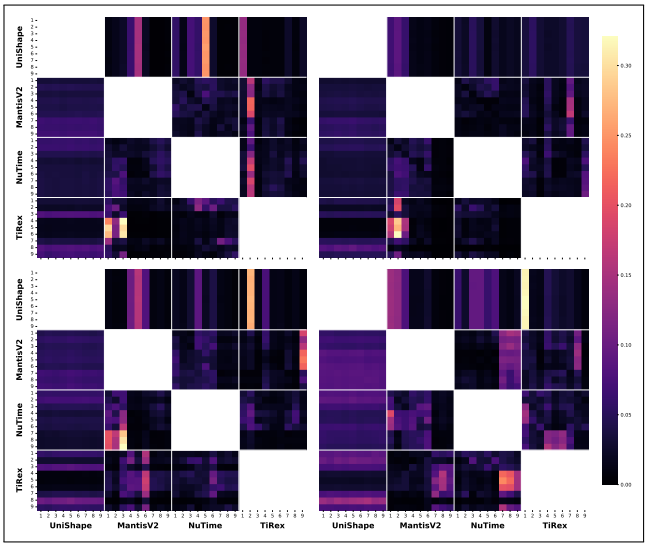


Figure 3: Cross-attention weights across the four heads of the EF (U+M+N+Ti) ensemble, averaged over the new holdout validation set. Querying models (y-axis) attend to the tokens of the remaining models (x-axis). Empty diagonal blocks indicate the exclusion of self-attention, strictly enforcing cross-model feature interactions among UniShape, MantisV2, NuTime, and TiRex.

3.2 Late Fusion Evaluation

As shown in Table 2, the Class-Weighted XGBoost ensemble achieved the highest performance on the held-out validation set, attaining an accuracy of 96.32% and a Macro F_1 score of 96.87%. Despite this strong performance, the high representational capacity of gradient-boosted decision trees increases the risk of overfitting to the user-specific characteristics and data distribution of the validation set, potentially compromising generalization to unseen users in the official test set. To maximize robustness and generalization, the Global Linear Fusion strategy was selected for generating the final predictions on the unlabeled official test set of the SHL Recognition Challenge 2026. Although it achieved a slightly lower validation accuracy of 94.66% and a Macro F_1 score of 95.18%, its reliance on only five global fusion weights results in a substantially lower model complexity, thereby reducing variance and mitigating the risk of overfitting to the validation distribution. Figure 4 illustrates the confusion matrix for the Global Linear Fusion model on the held-out validation set.

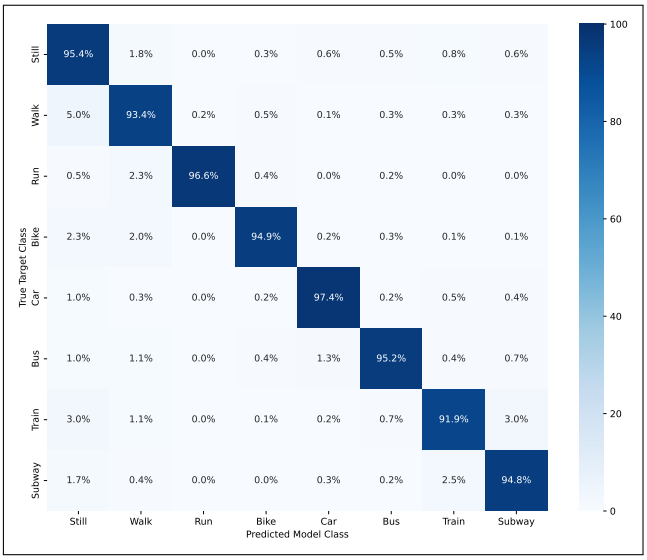


Figure 4: Confusion matrix for the Global Linear Fusion on the held-out Validation set.

The GPU-accelerated grid search allocated the optimal voting weights as follows: NuTime (0.54), Mantis (0.29), TiRex (0.17), UniShape (0.00), and TSPulse (0.00). The near-zero weights for TSPulse and UniShape indicate their final predictions are redundant with the stronger branches. This is likely because UniShape's shapelet-based predictions are subsumed by the spectral and statistical features specifically incorporated into the classification heads of other models. In contrast, the early fusion approach omitted these supplemental inputs and relied exclusively on FMs features, enabling UniShape to contribute essential, non-redundant representations.

3.3 Ensemble Robustness and Sensitivity Analysis

To rigorously validate the stability of the optimized global weights in our late fusion ensemble, we conducted a weight sensitivity analysis. We simulated 1,000 slightly altered weight configurations by introducing ±5% noise to the optimal parameters. The linear fusion demonstrated exceptional stability, maintaining a maximum accuracy of 94.682% and a minimum of 94.293%, with a negligible standard deviation of 0.051%. This confirms that the ensemble is situated in a wide, robust optimization minimum rather than a brittle local optimum.

Furthermore, to simulate severe sensor degradation or environmental interference, we injected escalating Gaussian noise directly into the model predictions.

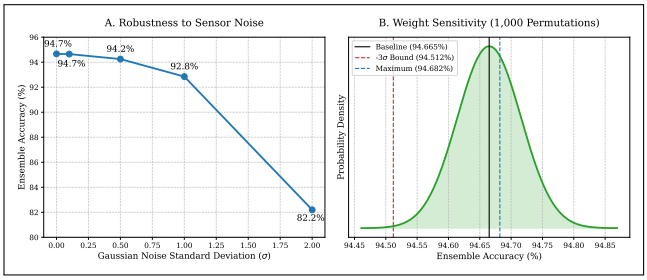


Figure 5: Impact of escalating Gaussian noise injection on the 3-model linear ensemble's validation accuracy.

As illustrated in Figure 5, the 3-model architecture proved highly resilient to moderate interference. Introducing a noise standard deviation of 0.1 and 0.5 resulted in accuracy drops of only 0.01% (to 94.65%) and 0.41% (to 94.25%), respectively. Even under severe noise injection (standard deviation of 1.0), the ensemble retained a 92.85% accuracy, experiencing a modest 1.81% drop. The system only began to significantly degrade at extreme noise levels (standard deviation of 2.0), falling 12.46% to an accuracy of 82.20%. These stress tests definitively prove that the lean, mathematically bounded linear fusion strategy operates as a highly robust filter against sensor unreliability in real-world locomotion tracking.

4 Conclusion

This paper presented early and late fusion frameworks integrating five frozen time-series foundation models (FMs) for IMU-based transportation mode recognition. Our early fusion employed a shared latent space with cross-model attention, while late fusion combined predictions via a class-weighted XGBoost meta-learner and a global linear strategy.

Our findings highlight key architectural trade-offs: (1) Model Families: Classification-native FMs (e.g., NuTime) achieve high accuracy but risk overfitting, whereas forecasting FMs (e.g., TiRex) provide robust sequential state-tracking. (2) Scalability & Channels: Global linear fusion easily scales to more activity classes, as its parameter count ($M = 5$) and search space remain class-independent, requiring only output dimension adjustments in the FM-specific heads. Additionally, independent FM channel processing enables seamless adaptation to arbitrary multi-sensor layouts. (3) Complexity: Our mathematically bounded global linear fusion reduces variance, compute overhead, and overfitting compared to high-capacity tree-based meta-learners.

Ultimately, fusing these FMs improved overall system robustness and recognition performance without end-to-end fine-tuning. We confirm all foundation model parameters remained strictly frozen throughout our pipeline. Final testing results will be detailed in the challenge summary paper [11].

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